optimization routines and computational power allow the required formulation to be solved quickly and efficiently. For example, see Malamut (2002) and www.KissellResearch.com.

The appropriate optimization technique is based on a multi-period process that segments the time horizon into a trading period where shares are transacted and an investment holding period where there are no further changes to the portfolio. With this new multi-period formulation it is possible that a suboptimal Markowitzian portfolio (e.g., below the efficient investment frontier) will result in better performance and higher utility due to more favorable trading statistics (liquidity and volatility). We refer to this set of ex-ante optimal portfolios as the best execution frontier.

To summarize, our main findings in this chapter are:

- There exist multiple sets of cost-adjusted frontiers for every efficient portfolio on the efficient investment frontier.
- There is a single "optimal trading strategy" that is consistent with the investment objective resulting in a single optimal cost-adjusted frontier. This is the best execution strategy for the investment portfolio.
- The proper level of risk aversion for a trade cost optimization to be consistent with the investment objective of the fund is the Sharpe ratio of the portfolio, or the forecasted Sharpe ratio of the investment decision.
- Evidence that a VWAP strategy is seldom consistent with the investment objectives and may lead to a suboptimal portfolio and lower levels of investor utility.
- A formulated multi-period investment portfolio optimization problem that considers both market impact cost and trading risk with the investment decision and leads to the best execution frontier.
- The formulated model provides opportunity to achieve cost reduction for a diversified trade list.
- The formulated model provides the preferred portfolio and corresponding road map (trade strategy) to build into those holdings.
- Market impact dominates ex-post performance much more than timing risk. Market impact results in a direct reduction in cost whereas timing risk is a sub-additive function and does not have the same linear relationship with portfolio risk.
- Post-trade analysis needs to incorporate the estimated costs of the trade (e.g., market impact and trading risk), and not solely rely on a benchmark price.